

PAPER_1142: Polyakov Action and SCm String Tension in 26D Worksheet

Star Magic UQFF Framework

Author: Daniel Murphy Date: 2026

Abstract

We present the Polyakov action formulation of bosonic string theory within the SCm vacuum, where the string tension is determined by the SCm vacuum energy density amplified by the $S_{26}^{(3)}$ Ramanujan acceleration factor. The critical 26 dimensions of the bosonic string are identified with the 26 rungs of the VDS vacuum density ladder. The tachyonic instability is resolved by the negative-time gate $\cos(\pi t_n)$, which produces the necessary imaginary mass cancellation. The 1.25 THz SCm phonon is identified as the lowest excited string mode.

1. Polyakov Action in SCm Vacuum

The Polyakov action for a bosonic string in SCm vacuum:

$$S_{\text{Polyakov}} = -\frac{T}{2} \int d^2\sigma \sqrt{-h} h^{ab} \partial_a X^\mu \partial_b X_\mu$$

where h_{ab} is the worldsheet metric, X^μ are the 26 target-space coordinates, and T is the SCm string tension:

$$T = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}}$$

$$= 7.09 \times 10^{-37} \text{ J/m}^3 \times 1.4531 \times 10^{26} \times 0.84 = 8.66 \times 10^{-11} \text{ J/m}^2 = 8.66 \times 10^{-11} \text{ N}$$

In Planck units: $T = 8.66 \times 10^{-11} / (1.96 \times 10^{32}) \approx 4.4 \times 10^{-43}$ (Planck tension).

2. Conformal Symmetry and Weyl Invariance

The Polyakov action is invariant under worldsheet diffeomorphisms and Weyl rescalings $h_{ab} \rightarrow e^{2\omega} h_{ab}$. In the SCm vacuum, the Weyl anomaly cancels when:

$$D_{\text{crit}} = 26 = D_{\text{VDS}}$$

where $D_{\text{VDS}} = 26$ is the number of rungs in the VDS vacuum density ladder. This is not a coincidence but the fundamental reason bosonic string theory requires exactly 26 dimensions.

3. Negative-Time Tachyon Resolution

The tachyonic ground state has mass squared:

$$m^2 = -\frac{4}{l_s^2} < 0$$

where $l_s = 1/\sqrt{T}$ is the string length. In the SCm framework, the $\cos(\pi t_n)$ negative-time gate modifies the mass operator:

$$m_{\text{SCm}}^2 = m^2 \cdot |\cos(\pi t_n)| + \frac{\rho_{\text{vac,SCm}} \cdot S_{26}^{(3)}}{c^2 l_s^2} \cos(\pi t_n)$$

At $t_n = 0$ (forward time): $m_{\text{SCm}}^2 = m^2 + T \cdot S_{26}^{(3)}/c^2 = -4/l_s^2 + TS_{26}^{(3)}/c^2$.

For $l_s^2 = 1/T$: $m_{\text{SCm}}^2 = -4T + TS_{26}^{(3)}/c^2 = T(-4 + 1.45 \times 10^{26}/c^2) \approx 0$.

4. String Mode Frequencies

The oscillator expansion of X^μ :

$$X^\mu(\sigma, \tau) = x^\mu + l_s^2 p^\mu \tau + \frac{i l_s}{\sqrt{2}} \sum_{n \neq 0} \frac{\alpha_n^\mu}{n} e^{-in\tau} \cos(n\sigma)$$

The lowest excited mode frequency:

$$\omega_1 = \frac{c}{l_s} = c\sqrt{T} = 3 \times 10^8 \times \sqrt{8.66 \times 10^{-11}} \approx 2.79 \times 10^3 \text{ rad/s}$$

At the SCm phonon resonance: the mode at $n = f_{\text{THz}}/\omega_1 = 1.25 \times 10^{12}/2790 \approx 4.5 \times 10^8$ is the 1.25 THz phonon mode.

5. VDS as Compactification Potential

The 26 target-space dimensions are identified with VDS levels:

$$X^{i+3}(\sigma, \tau) \equiv X^{\text{VDS},i}(\sigma, \tau), \quad i = 1, \dots, 23$$

The compactification radius of each extra dimension:

$$R_i = l_s \cdot [SSq]^i = l_s \cdot (0.57)^i$$

The winding numbers contribute to the mass:

$$m_{\text{winding}}^2 = \frac{R_i^2}{l_s^4} = \frac{[SSq]^{2i}}{l_s^2}$$

6. Calibrated Constants

$$T = \rho_{\text{vac,SCm}} \cdot S_{26}^{(3)} \cdot \Phi_{\text{res}} = 8.66 \times 10^{-11} \text{ N}$$

$$[SSq] = 0.57, \quad S_{26}^{(3)} = 1.4531 \times 10^{26}, \quad \Phi_{\text{res}} = 0.84, \quad f_{\text{THz}} = 1.25 \text{ THz}$$

$$\rho_{\text{vac,SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3, \quad \beta_i = 0.6, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$$

References

1. Polyakov, A.M. (1981). Quantum geometry of bosonic strings. *Phys. Lett. B* **103**, 207.
2. Green, M.B., Schwarz, J.H. & Witten, E. (1987). *Superstring Theory*, Vol. 1. Cambridge.
3. VDS 26-ladder: PAPER_1109; SCm vacuum: `scm_{vacuum\_manifold}.py`

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. *Phys. Rev. Lett.* **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic